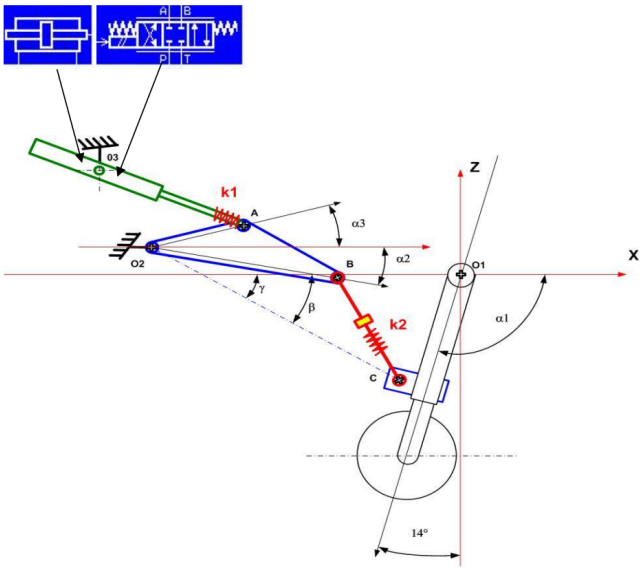
Institut Polytechnique des Sciences Avancées (IPSA)

# Modeling and Simulation of a Hydraulic Landing Gear

*Multi-Physics Analysis using Simcenter Amesim*



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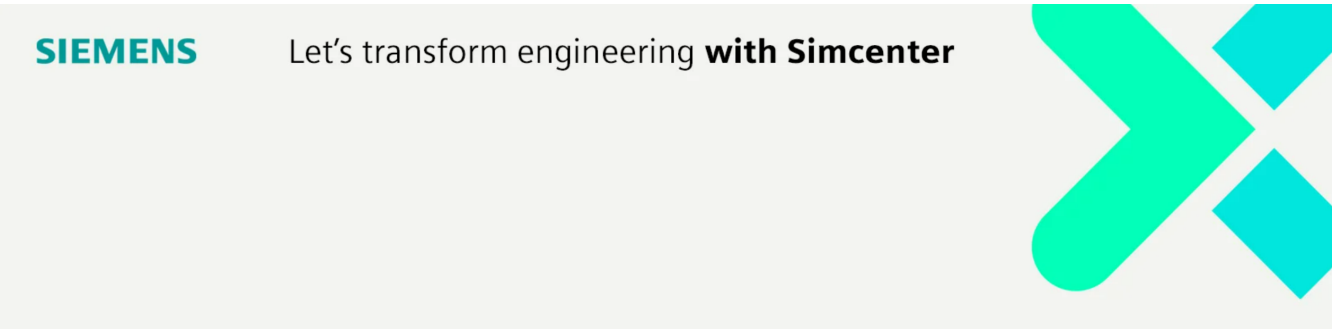
# Introduction

This project focuses on the design and analysis of a landing gear actuation system. The primary objective is to transition from a complex kinematic diagram to a dynamic simulation model using **Simcenter Amesim**.

The main challenge lies in understanding the power transmission between three distinct physical domains: **Hydraulic** (power source), **Mechanical Translation** (actuator), and **Mechanical Rotation** (deployment mechanism). This report covers the structural modeling, parameterization, and open-loop performance analysis.




# 1 System Study

The structural analysis is based on decomposing the landing gear into functional sub-blocks linked by power variables (Effort and Flow). This approach follows the Bond Graph theory, where the product of these variables represents the instantaneous power exchange between domains.

## 1.1 Architecture and Power Flow

The system is driven by a high-pressure hydraulic source delivering a constant supply pressure of 206 bar. The power flow, representing the energy transmission from the pilot's command to the final mechanical deployment, follows this specific path:

1. **Servovalve**: Acts as a modulated resistance ($R$). It regulates the fluid flow ($Q$) and the pressure differential ($\Delta P$) sent to the chambers based on the electrical input signal.
2. **Actuator (Cylinder)**: Functions as a multi-port transformer ($TF$). It converts hydraulic power ($P \cdot Q$) into mechanical translation power ($F \cdot v$). The conversion is governed by the piston's effective surface area.
3. **Blue Triangle ($O_2AB$)**: This mechanical component serves as a rotational transformer ($TF$). It converts the linear motion from the actuator into an angular velocity ($\omega$) and a torque ($\tau$) through its specific geometry and lever arm ratios.
4. **Linkage ($BC$)**: A critical interface that transmits the force to the landing gear leg. It is modeled as a compliant rod to account for the structural elasticity under high loads.
5. **Landing Gear Leg**: The final stage of the power chain, which performs the rotation around the main pivot $O_1$. This stage is where the actual deployment work is performed against gravity and aerodynamic forces.

## 1.2 Identification of R, I, C Elements

To accurately simulate the dynamic behavior, each physical domain was analyzed to identify its fundamental energy properties (Dissipation, Inertia, or Storage):

- **R (Resistance - Energy Dissipation)**:
  - Major pressure drops occur within the servovalve ($\Delta P_{nom} = 100$ bar at rated flow).
  - Secondary dissipation is caused by viscous friction in the hydraulic lines and mechanical friction at the pivots ($O_1, O_2$).
- **I (Inertia - Kinetic Energy Storage)**:
  - The linear inertia is represented by the piston and rod mass (50 kg).

  - The rotational inertia accounts for the heavy structure of the triangle and the leg assembly ($290\,\mathrm{kg}\cdot\mathrm{m}^2$), which limits the acceleration of the deployment.

- **C (Capacity - Potential/Elastic Energy Storage)**:
  - The fluid's bulk modulus accounts for hydraulic compressibility.
  - The structural stiffness of the linkages ($k_1, k_2 = 2.1 \times 10^7\,\mathrm{N/m}$) allows for energy storage and determines the system's resonance frequencies.

## 1.3 Transformation Laws

The mechanical deployment relies on the principle of lever arms. The blue triangle and the leg assembly are modeled as transformers ($TF$) where the transmission ratio is defined by the distance between the application point and the pivot axis:

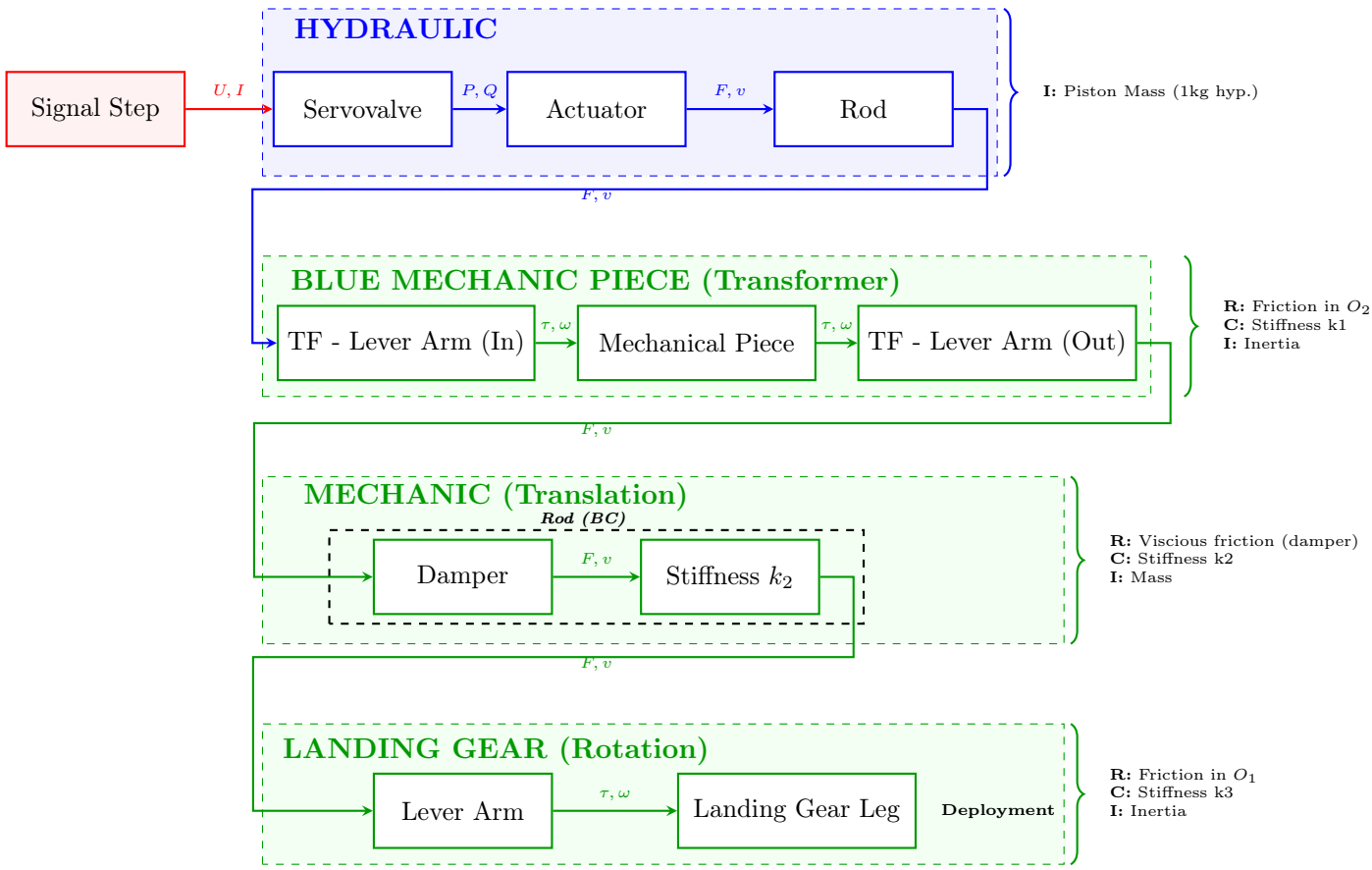
The fundamental relations linking the linear and rotational domains are:

$$\tau = F \cdot L \quad \text{and} \quad v = \omega \cdot L$$

For the blue triangle, two distinct lengths $L$ are considered: 1000 mm for the input point $A$ (connection to the actuator) and 3500 mm for the output point $B$ (connection to the linkage). This difference results in a significant torque multiplication and speed adaptation across the mechanism.

## 1.4 Block diagram

Now, let's represent the landing gear system into a block diagram to visualize the power transfert, efforts and flows :



### 1.5 Full Power Chain Decomposition

The system is modeled as a series of power transducers. Following the Bond Graph formalism, we define the instantaneous power as $P(t) = e(t) \cdot f(t)$, where $e$ is the **Effort** and $f$ is the **Flow**.

1. **MODULE: HYDRAULIC (Electro-Hydraulic Conversion)**
   - **Input Power ($P_{in,H}$):**
   $$P_{in,H} = U \cdot I$$
   - **Nature:** Electrical signal (Command).
   - **Internal Process:** The Servovalve acts as a modulated resistance ($R$), creating a pressure drop $\Delta p$. The Actuator ($TF$) then converts this hydraulic energy into mechanical power.
   - **Output Power ($P_{out,H}$):**
   $$P_{out,H} = F_{cyl} \cdot v_{cyl}$$
   - **Nature:** Linear Mechanical (Translation).
   - *Note:* $P_{out,H} = (P_{chamber} \cdot A_{piston}) \cdot v_{piston}$.
2. **MODULE: BLUE MECHANIC PIECE (Mechanical Transformation)**
   - **Input Power ($P_{in,M1}$):**
   $$P_{in,M1} = P_{out,H} = F_{cyl} \cdot v_{cyl}$$
   - **Internal Process:**
     - **Lever Arm In:** Converts $F_{cyl} \cdot v_{cyl}$ into $\tau_A \cdot \omega_{O2}$.
     - **Inertia ($J$):** Stores energy as kinetic power $P_J = J \cdot \omega \cdot \dot{\omega}$.
     - **Lever Arm Out:** Converts $\tau_B \cdot \omega_{O2}$ back into linear power for the rod.
   - **Output Power ($P_{out,M1}$):**
   $$P_{out,M1} = F_{rod,in} \cdot v_{rod,in}$$
   - **Nature:** Linear Mechanical (Translation).
3. **MODULE: MECHANIC TRANSLATION (Elastic Linkage BC)**
   - **Input Power ($P_{in,T}$):**
   $$P_{in,T} = P_{out,M1} = F_{rod,in} \cdot v_{rod,in}$$
   - **Internal Process:** Energy is partially stored in the stiffness $k_2$ and dissipated in the damper.
     - $P_{dissipated} = R \cdot v^2$
     - $P_{stored} = \frac{d}{dt}\left(\frac{1}{2}k_2 x^2\right)$
   - **Output Power ($P_{out,T}$):**
   $$P_{out,T} = F_{rod,out} \cdot v_{rod,out}$$
   - **Nature:** Linear Mechanical (Translation).
4. **MODULE: TRAIN (Final Actuation)**
   - **Input Power ($P_{in,L}$):**
   $$P_{in,L} = P_{out,T} = F_{rod,out} \cdot v_{rod,out}$$
   - **Internal Process:** The final Lever Arm ($TF$) transforms the rod's linear power into the Leg's rotational power.
   - **Output Power ($P_{out,L}$):**
   $$P_{out,L} = \tau_{leg} \cdot \omega_{O1}$$
   - **Nature:** Rotational Mechanical (Deployment).
   - **Final Work:** This power is used to overcome gravity ($mgh$) and aerodynamic drag ($P_{aero}$).

Table 1: Power Conversion Summary Table

| Sub-module | Effort Variable ($e$) | Flow Variable ($f$) | Power Nature |
|---|---|---|---|
| **Hydraulic** | Pressure $P$ | Vol. Flow $Q$ | Elec → Hydro → Trans. |
| **Transformer** | Force $F_{cyl}$ | Velocity $v_{cyl}$ | Trans → Rot → Trans. |
| **Translation** | Force $F_{rod,in}$ | Velocity $v_{rod,in}$ | Trans (Elastic Linkage) |
| **Train** | Force $F_{rod,out}$ | Velocity $v_{rod,out}$ | Trans → Rot. |

# 2 AMESIM Simulation Model - OPEN LOOP

Following the theoretical analysis and the establishment of the power flow chain, the system was implemented in **Simcenter Amesim**. This environment allows for a 1D multi-physics simulation where each mechanical and hydraulic phenomenon is represented by a specific validated sub-model. The transition to Amesim is crucial to validate the dynamic behavior of the landing gear, particularly the influence of structural stiffness and hydraulic damping on the deployment time.

## 2.1 Model Construction and Library Elements

### 2.1.1 Hydraulic Section

The hydraulic circuit is designed to transition from the control signal to mechanical power. Each component selected from the **Amesim Hydraulic Library** plays a specific role in the energy modulation chain.

**Control Signal Source (Step)** The simulation begins with a **Signal Step** block, which generates the pilot's command in the form of an electrical current (mA). This element represents the input interface of the system, defining the timing and magnitude of the deployment sequence. It acts as the primary driver for the downstream multi-physics transitions.

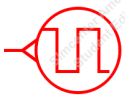


Figure 1: Signal Step (Signal02)

**Servovalve (4-Way Spool Valve)** The **4-way Servovalve** is the central resistive element ($R$) of the hydraulic domain. It modulates the high-pressure fluid flow ($Q$) into the cylinder chambers based on the input signal. By creating a controlled pressure drop ($\Delta p$), it regulates the speed of the actuator and ensures a smooth power distribution between the supply and the return lines.

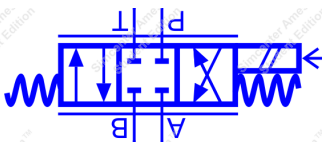


Figure 2: Servovalve (hsv 3pos4ports 01)

**Hydraulic Actuator (Cylinder)** The **Double-acting Cylinder** serves as the critical Transformer ($TF$) between the hydraulic and mechanical domains. It converts the fluid's static pressure ($P$) and volumetric flow ($Q$) into a linear mechanical force ($F$) and velocity ($v$). This component accounts for the piston's effective surface area and represents the physical interface with the landing gear's mechanical linkage.

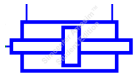


Figure 3: Actuator (actuator002)

**Power Unit (Pressure Source and Tank)** The hydraulic power is provided by a **Constant Pressure Source** and a **Tank** model. The source simulates the aircraft's central hydraulic system, typically maintained at 206 bar (3000 psi), while the tank represents the low-pressure return reservoir. Together, they establish the boundary conditions necessary for the fluid circulation and pressure differentials within the circuit.

Figure 4: Pressure source and tank (pressuresource and tank01)

At the end of the day, we obtain the following hydraulic subpart:

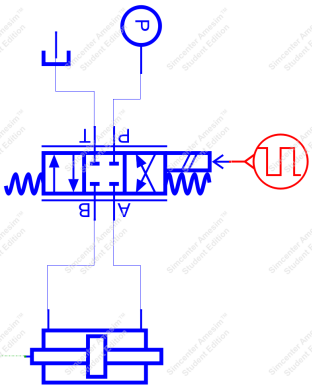


Figure 5: Hydraulic subpart

## 2.2 Blue Mechanic Piece Section

The transformation of translational motion into rotational motion is managed by the **Planar Mechanical Library**. This section represents the rigid "Blue Triangle" linkage which dictates the kinematic behavior of the landing gear deployment.

**Auxiliary Mass for Causality (M)** A **Mechanical Mass** block (violet) was intentionally inserted between the hydraulic actuator and the structural spring. In Bond Graph theory, both the hydraulic cylinder and the mechanical spring are defined as **Effort Sources** (or Capacitive elements). Connecting them directly would result in a causality conflict, as the solver cannot compute the velocity without an intermediate **Inertial element** ($I$). This mass is introduced to break this conflict, allowing the calculation of acceleration via Newton's Second Law. To ensure the system's dynamic response remains unaffected, this parameter will later be set to a negligible value of 1 kg.

Figure 6: Mass (causality)

**Lever Arm Transformers (TF)** The "Blue Piece" is modeled using two **Lever Arm** components connected to a common pivot. These blocks function as Transformers ($TF$), converting the linear force ($F$) from the actuator into torque ($\tau$) for the first arm, and then converting the angular velocity ($\omega$) back into linear motion for the second arm. This dual-arm configuration allows the model to respect the specific geometry of the triangle, where different lengths $L_1$ and $L_2$ are used to adapt the mechanical advantage.

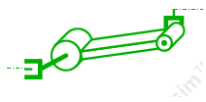


Figure 7: Lever Arm

**Rotational Inertia (J)** At the center of the two arms, a **Rotational Inertia** block represents the mass distribution of the triangle relative to its fixed pivot $O_2$. In this Amesim configuration, this block also acts as the mechanical ground for the rotation, defining the fixed axis around which the triangle oscillates. It is crucial for capturing the transient phases of the deployment, as it resists sudden changes in angular velocity according to the moment of inertia $J_{triangle}$.

Figure 8: Inertia

Here is the full section on AMESIM:

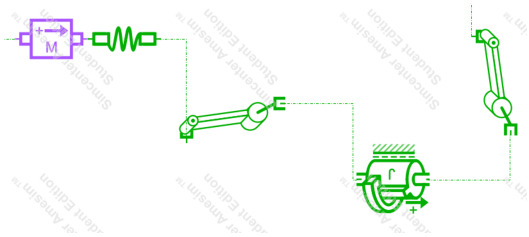


Figure 9: Full Blue Mechanical Piece Section

## 2.3 Mechanical Translation (Connecting Rod BC)

The connecting rod, situated between the blue triangle and the main landing gear leg, is modeled as a translational assembly. This section captures the structural dynamics and power transmission through the linkage.

**Spring-Damper Assembly** The rod is represented by a combined **Spring-Damper** component from the mechanical library. This element incorporates both a structural stiffness ($k_2$) and a viscous damping coefficient ($R$). In the Bond Graph approach, the spring acts as a **Capacitive element** ($C$), storing elastic potential energy during high-load phases, while the damper acts as a **Resistive element** ($R$), dissipating energy to suppress unwanted structural oscillations. This modeling choice is essential for simulating the realistic "flex" of the linkage under hydraulic pressure.

Figure 10: Spring damper

## 2.4 Landing Gear Assembly (Leg Rotation)

The final stage of the Amesim model represents the landing gear leg's rotational deployment around the main aircraft pivot $O_1$. This assembly converts the translational effort from the BC rod into the final angular movement.

**Final Lever Arm (Arm)** A translational-to-rotational **Arm** component acts as the final Transformer ($TF$) of the power chain. It receives the linear force and velocity from the connecting rod and converts them into torque ($\tau$) and angular velocity ($\omega$). This block is parameterized by the distance between the pivot $O_1$ and the attachment point of the rod, determining the mechanical advantage required to lift the leg's weight against gravity.

**Landing Gear Inertia (J)** The main structural mass of the landing gear is modeled by a **Rotational Inertia** block. This represents the cumulative inertia $J_{leg}$ of the leg, wheel, and tire assembly. This inertial element ($I$) is the primary load for the hydraulic system, determining the acceleration profile and the time required for full deployment. It also serves as the reference for the angular position of the gear, which is the main output variable of the simulation.

**Angular Reference (Zero Torque Source)** A **Zero Torque Source** is used at the end of the mechanical chain to provide a consistent numerical reference for the rotational ports. In Bond Graph terms, this component acts as a neutral boundary condition that satisfies the solver's causality requirements without introducing any parasitic resistance or external load. By maintaining a torque value of zero, it allows the landing gear to pivot freely under the influence of the primary actuator and the leg's inertia, ensuring that the simulated motion is governed solely by the system's internal dynamics and geometric constraints.

Figure 11: Zero Torque Source

In conclusion, here is the full AMESIM model, linking all subsections together :

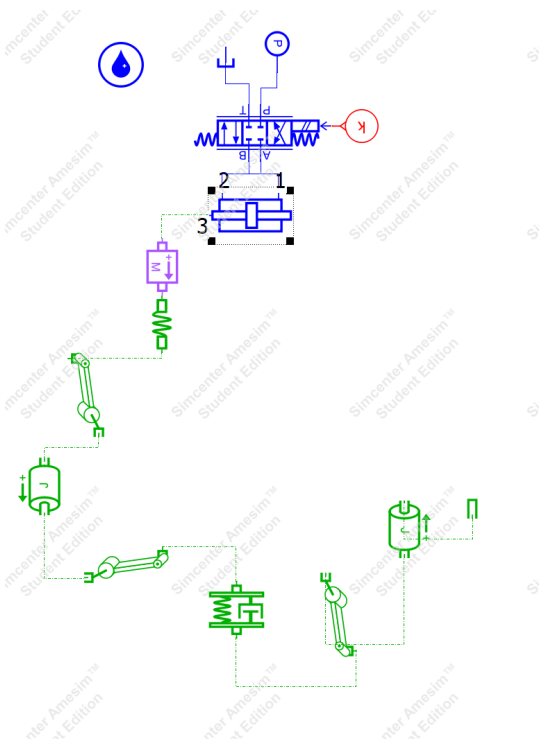



Figure 12: Full AMESIM model

# 3 System Parameterization - OPEN LOOP

Once the structural architecture of the model is validated, numerical values must be assigned to each component to reflect the physical properties of a real aeronautical landing gear. The parameterization phase is critical as it defines the time constants, the system's damping, and the final force-torque balance. These values are based on standard aeronautical data for regional aircraft (e.g., 3000 psi hydraulic systems and high-stiffness structural linkages).

**Numerical Data Specifications**

**Mechanical Properties**

- ALL Inertias: $290\,\mathrm{Kgm}^2$
- ALL Stiffness rotation: $5000\,\mathrm{Nm}/^\circ$
- ALL Masses: $50\,\mathrm{Kg}$
- ALL Stiffness linear: $2.1 \times 10^7$

**Lever Arm Geometry**

- A-O2: 1000 mm
- O2-B: 3500 mm
- B-C: 2000 mm

**Actuator & Valve**

- Static Max forces: 25 000 N
- Max velocity: 800 mm/s
- Max stroke: 1000 mm
- Nom. press. drop/orifice: 100 bar
- Supply pressure: 206 bar

## 3.1 Signal Section

### Control Input Signal (Step Command)

- **Signal Type:** Unit Step / Constant (1.0)
- **Amplitude:** 1 (representing 100% of the servovalve's rated current)

  *Justification:* A constant unit signal is used to simulate a full deployment command. In a landing gear system, the servovalve must be fully opened to allow the maximum flow rate (60 L/min) to reach the actuator. Given the high rotational inertia of the assembly ($290\,\mathrm{kg} \cdot \mathrm{m}^2$), a steady command is necessary to overcome the initial static resistance and ensure the piston reaches its target stroke of 1000 mm within a realistic operational timeframe. This choice provides a clear baseline for analyzing the system's step response and transient stability.

## 3.2 Hydraulic Section Parameters

This subsection lists the critical values for the power generation and modulation elements. The goal is to establish a steady-state supply that allows for a fast and stable deployment.

### Hydraulic Fluid Properties

- **Fluid Index:** 0
- **Density ($870\,\mathrm{kg/m}^3$):** This value is set to match the standard properties of aeronautical hydraulic oil (MIL-H-5606 type).
- **Bulk Modulus (17000 bar):** This high value represents the fluid's resistance to compression, which is essential for the rapid transmission of pressure waves in the landing gear system.

  *Justification:* The inclusion of the Hydraulic Fluid Property block is a fundamental modeling requirement in Simcenter Amesim. It serves as a global database that provides the physical characteristics (viscosity, density, and compressibility) to all hydraulic components sharing the same fluid index. Using realistic values for an "unnamed fluid" ensures that the pressure drops in the servovalve and the force transmission in the actuator are physically accurate.

### Pressure Source (PS00-1)

- **Pressure at start of stage 1:** 206 bar
- **Pressure at end of stage 1:** 206 bar
- **Duration of stage 1:** $1.0 \times 10^6$ s

*Justification:* The supply pressure is set to 206 bar (3000 psi), which is the standard operating pressure for civil aircraft hydraulic power units. A long duration is chosen to ensure constant pressure throughout the entire simulation transient.

### Hydraulic Tank

- **Tank Pressure:** 1 bar
- **Liquid Density:** $870\,\mathrm{kg/m}^3$ (Standard Hydraulic Oil)

*Justification:* The tank pressure is set to 1 bar to simulate a return to atmospheric pressure. This creates the maximum pressure differential ($\Delta P = 205\,\text{bar}$) across the system, ensuring the required flow rate for the deployment while preventing numerical cavitation in the return lines.

### Servovalve Flow Characteristics

- **Characteristic flow rate (60 L/min):** This value is calculated to match the actuator's performance requirements.

- **Corresponding pressure drop (100 bar):** Taken directly from the technical specifications.

  *Justification:* To achieve the required maximum velocity of 800 mm/s with a 40 mm piston, the flow rate $Q$ must satisfy $Q = v \cdot A_p$. Calculating $800\,\text{mm/s} \times 1256.6\,\text{mm}^2$ yields approximately 60 L/min ($Q = v_{max} \cdot A_p = 800\,\text{mm/s} \cdot 1256.6\,\text{mm}^2 = 1\,005\,280\,\text{mm}^3/\text{s}$; After converting in L/min we obtain :

  $$Q \approx 60.3\,\text{L/min}$$

  Setting this flow rate at the nominal 100 bar pressure drop ensures the servovalve is correctly sized to drive the landing gear within the specified dynamic limits without flow saturation.

### Hydraulic Actuator (HC000)

- **Piston and Rod Diameters (40 mm/12 mm):** These dimensions are determined by the system's force requirements and supply pressure.

  *1. Piston Sizing:* To meet the static force requirement of $F_{static} = 25\,\text{kN}$ under a supply pressure of 206 bar (20.6 MPa), and knowing that $F_{req} = P_s \times A_p = P_s \times \frac{\pi \cdot D_p^2}{4}$, the required piston diameter $D_p$ is calculated as follows:

  $$D_p = \sqrt{\frac{4 \cdot F_{static}}{\pi \cdot P_s}} = \sqrt{\frac{4 \cdot 25,000}{\pi \cdot 20.6 \cdot 10^6}} \approx 39.2\,\text{mm}$$

  Consequently, a standard diameter of 40 **mm** is selected to provide a safety margin and ensure the gear can be deployed even with internal friction.

  *2. Rod Sizing:* The rod diameter $d_r = 12\,\text{mm}$ is chosen based on structural requirements. While the piston area $A_p \approx 1256.6\,\text{mm}^2$ governs the output force, the rod diameter defines the annular area $A_{ann} = A_p - A_{rod} \approx 1143.5\,\text{mm}^2$. This configuration results in a speed ratio of:

  $$\frac{v_{in}}{v_{out}} = \frac{A_p}{A_{ann}} \approx 1.1$$

  This 10% difference between retraction and deployment speeds ensures a balanced hydraulic behavior, while the 12 mm steel rod provides the necessary resistance against buckling under the maximum 25 kN load (choosing 20mm, so considering a coefficient of 0.5 between the size of the piston and the size of the rod, we could have a 25% difference, which is too important).

- **Length of Stroke (1000 mm):** This parameter is strictly defined by the technical specifications of the actuator. A maximum stroke of 1000 mm (or 1 m) is chosen to ensure full kinematic deployment of the landing gear assembly. This value provides a sufficient operational range to perform the required 500 mm step response test without reaching the mechanical end-stops prematurely.

- **Dead Volumes ($50\,\text{cm}^3$):** Since the exact dead volumes are not specified in the numerical data, an assumption must be made. A volume of $50\,\text{cm}^3$ is set for both ends to model the hydraulic capacitance ($C$). Mathematically, this prevents the fluid volume from reaching absolute zero at the ends of the stroke, which would cause an infinite hydraulic stiffness and lead to a numerical singularity (solver divergence).

- **Viscous Friction and Leakage Coefficients (0):** For this initial open-loop analysis, the actuator is assumed to be in an ideal state. The primary energy dissipation ($R$) is already governed by the servovalve's nominal pressure drop (100 bar per orifice) and the mechanical structural damping. Setting internal leakage and viscous friction to zero simplifies the validation of the fundamental power transfer and force calculations.

- **Endstops Stiffness and Damping ($10^5$):** The actuator has a maximum stroke of 1000 mm. To simulate the mechanical limits at full retraction and full deployment, highly

rigid endstops are required. A spring rate of $100,000\,\mathrm{N/mm}$ and an equivalent high damping coefficient ensure that the piston stops abruptly without experiencing unphysical numerical bouncing (chattering) upon impact with the cylinder head.

## 3.3 Mechanic Section Parameters

### Auxiliary Causality Mass

- **Mass ($1\,\mathbf{kg}$):** This component is a numerical necessity introduced to break the causality conflict between the hydraulic actuator and the mechanical spring.

  *Justification:* In Simcenter Amesim, connecting two components that both impose an effort (displacement/force) without an intermediate element leads to a solver error. A mass of $1\,\mathrm{kg}$ is selected to satisfy the software's causality requirements while ensuring that this added inertia is negligible compared to the system's total mass ($50\,\mathrm{kg}$) and rotational inertia ($290\,\mathrm{kg}\cdot\mathrm{m}^2$). This ensures that the dynamic response and the natural frequency of the landing gear deployment remain physically accurate and unaffected by this modeling artifact.

### Linear Spring (k1)

- **Stiffness ($2.1\times10^7\,\mathbf{N/m}$):** This value is taken directly from the numerical data.*Justification:* It represents the structural rigidity of the mechanical linkage. This high stiffness acts as a capacitive element ($C$) in the power chain, allowing for the storage of elastic potential energy and ensuring a realistic transmission of forces between the actuator and the gear assembly.

### Lever Arm 1 (AO2)

- **Length ($1000\,\mathbf{mm}$):** This parameter is taken directly from the project's numerical data.

  *Justification:* This length defines the first transformer ratio ($TF$) of the mechanical assembly. It determines how the linear force from the hydraulic actuator is converted into torque around the pivot $O_2$. Maintaining this exact dimension is essential to ensure the correct mechanical advantage required to initiate the rotation of the landing gear.

### Rotary Load (Pivot O2)

- **Moment of Inertia ($290\,\mathbf{kg\cdot m^2}$):** This value is taken from the "ALL Inertias" specification in the project data.

  *Justification:* This parameter represents the rotational inertia of the triangular bracket assembly. In the power chain, it acts as an **Inertial element ($I$)** that opposes changes in angular velocity. Accurately modeling this inertia is critical for determining the dynamic response and the torque required from the actuator to accelerate the mechanism during the deployment transient.

### Lever Arm 2 (O2B)

- **Length ($3500\,\mathbf{mm}$):** This value is defined by the geometry of the triangular bracket.

  *Justification:* This second lever arm acts as a transformer ($TF$) that dictates the motion transfer from the intermediate pivot $O_2$ to the connecting rod BC. The ratio between $AO_2$ ($1000\,\mathrm{mm}$) and $O_2B$ ($3500\,\mathrm{mm}$) provides a mechanical displacement amplification, which is a key characteristic of this landing gear's kinematics.

### Spring-Damper BC (Linkage)

- **Spring Rate ($2.1\times10^7\,\mathbf{N/m}$):** This value is taken from the numerical data to model structural stiffness.

- **Damper Rating ($1000\,\mathbf{N/(m/s)}$):** Since no damping value was provided in the specifications, an assumption of $1000\,\mathrm{N/(m/s)}$ is made.

  *Justification:* This coefficient is necessary to provide viscous dissipation within the mechanical linkage. It ensures numerical stability by damping the high-frequency oscillations that could be triggered by the high spring rate and the large system inertias ($290\,\mathrm{kg}\cdot\mathrm{m}^2$). This value is chosen to allow a realistic transient response without overdamping the landing gear's movement.

### Lever Arm 3 (BC)

- **Length ($2000\,\mathbf{mm}$):** This value is taken from the "ALL Liver Arm" data in the project description.

*Justification:* This component represents the final mechanical transformer ($TF$) of the landing gear assembly. It converts the linear displacement and force from the connecting rod into the final angular rotation of the main leg around pivot $O_1$. The choice of 2000 mm defines the torque authority available to move the heavy landing gear assembly against gravity and aerodynamic loads.

**Landing Gear Inertia (Pivot O1)**

- **Moment of Inertia ($290\,\mathbf{kg \cdot m^2}$):** This fundamental parameter is taken from the project's numerical specifications.

  *Justification:* This value represents the total rotational inertia of the main landing gear leg, including the wheel and tire assembly. As the primary inertial load ($I$) in the power chain, it determines the torque required for acceleration and the overall deployment time.

- **Angular Endstops ($-5°$ to $100°$):** The angular limits have been set to cover the full kinematic range of the landing gear.

  *Justification:* A range from $-5°$ to $100°$ is selected to ensure that the simulation can reach the fully deployed state (typically 90°) without reaching mechanical saturation. The 5° margin at each end prevents numerical instability at the start and end of the motion, allowing the piston to reach its own mechanical stops (1000 mm) naturally.

## 3.4 Summary of System Parameters

The following tables synthesize the physical and numerical parameters implemented in the Simcenter Amesim model.

### 3.4.1 Hydraulic Domain Parameters

| Component | Parameter | Value | Source/Justification |
|---|---|---|---|
| **Pressure Source** | Supply Pressure | 206 bar | Standard Aero |
| **Hydraulic Tank** | Return Pressure | 1 bar | Atmospheric Ref. |
| **Servovalve** | Rated Current | 40 mA | Industrial Std. |
| | Press. Drop | 100 bar | Power Optimization |
| **Actuator** | Piston Diameter | 40 mm | Calculated ($F_{static}$) |
| | Rod Diameter | 12 mm | 10% Velocity Ratio |
| | Max Stroke | 1000 mm | Subject Data |
| | Dead Volumes | $50\,\mathrm{cm}^3$ | Numerical Stability |

Table 2: Consolidated hydraulic parameters.

### 3.4.2 Mechanical Domain Parameters

| Component | Parameter | Value | Source/Justification |
|---|---|---|---|
| **Causality Mass** | Auxiliary Mass | 1 kg | Numerical Necessity |
| **Linear Spring** | Stiffness ($k_1$) | $2.1 \times 10^7$ N/m | Subject Data |
| **Lever Arms** | $A - O_2$ / $O_2 - B$ / $B - C$ | 1000/3500/2000 mm | Kinematic Data |
| **Inertias** | Pivot $O_2$ (Bracket) | $1\,\mathrm{kg \cdot m^2}$ | Interface Inertia |
| | Pivot $O_1$ (Leg) | $290\,\mathrm{kg \cdot m^2}$ | Main Load |
| **Spring-Damper** | Damper Rating | 1000 N/(m/s) | Assumption |

Table 3: Consolidated mechanical and structural parameters.

## 3.5 Kinematic Analysis and Operational Constraints

The mechanical behavior of the landing gear is governed by the geometric ratios of the bell-crank linkage. This section demonstrates the theoretical maximum stroke and the subsequent implementation of the specific 14° vertical alignment requirement.

### 3.5.1 Theoretical Maximum Stroke: Geometric Singularity

Preliminary simulation results indicate that the actuator displacement stabilizes at $x_{piston} = 571.4\,\text{mm}$, despite a theoretical maximum cylinder stroke of 1000 mm. A step-by-step kinematic analysis confirms that this value corresponds to the full mathematical deployment of the linkage:

1. **Bracket Rotation ($O_2$):** The piston acts on the input lever arm ($L_1 = 1000\,\text{mm}$). The angular deflection is:

$$\theta_{bracket} = \arcsin\left(\frac{x_{piston}}{L_1}\right) = \arcsin\left(\frac{571.4}{1000}\right) \approx 34.85^\circ \tag{1}$$

2. **Linkage Displacement:** This rotation is transmitted via the output arm ($L_2 = 3500\,\text{mm}$) to the connecting rod interface:

$$x_{rod} = L_2 \times \sin(\theta_{bracket}) = 3500 \times \sin(34.85^\circ) \approx 2000\,\text{mm} \tag{2}$$

3. **Leg Deployment ($O_1$):** Given the final lever arm $L_3 = 2000\,\text{mm}$, the deployment angle reaches:

$$\theta_{leg} = \arcsin\left(\frac{x_{rod}}{L_3}\right) = \arcsin\left(\frac{2000}{2000}\right) = 90^\circ \tag{3}$$

At this point, a **geometric singularity** occurs: the final lever arm and the connecting rod are perfectly aligned. Any further stroke is physically impossible as it would require $\sin\theta > 1$.

### 3.5.2 Operational Limit: 14° Vertical Alignment

Per the technical specifications, the landing gear must not reach the singular 90° position but must lock at 14° relative to the vertical axis ($Z$). This constraint was integrated into the Amesim model to reflect the actual operational range:

- **Angular Target:** To maintain a 14° tilt from the vertical, the total angular travel from the retracted horizontal position is restricted to 76° (90° − 14°).
- **Model Implementation:** The *Ideal Endstop* parameters within the final rotational inertia submodel were configured with an upper limit of 76 deg.
- **Impact on Displacement:** Under this constraint, the piston stroke is reduced from the singular 571.4 mm to approximately 553 mm.

### 3.6 Results and interpretations

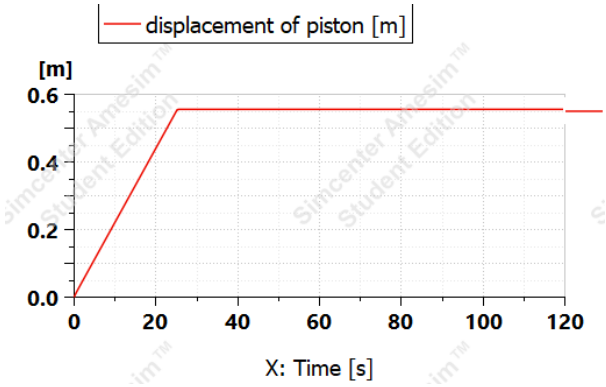



Figure 13: Piston displacement vs. Time

**Interpretation** The recorded displacement curve for the hydraulic actuator exhibits a linear transit phase ($0 < t < 25\,\mathrm{s}$) followed by an abrupt stabilization at $t \approx 25\,\mathrm{s}$.
During the initial phase, the displacement increases with a constant slope, indicating a steady-state velocity of approximately $0.023\,\mathrm{m/s}$. This linearity confirms that the flow rate delivered by the servovalve is stable and that the mechanical load is driven without significant oscillations.
The motion subsequently saturates at $x_{piston} \approx 0.553\,\mathrm{m}$, a stabilization point that corresponds to the mechanical end-of-travel defined by the 14° vertical alignment constraint. At this instant, the hydraulic energy is no longer converted into kinetic energy but is instead transformed into static pressure within the actuator chamber, marking the completion of the deployment sequence.

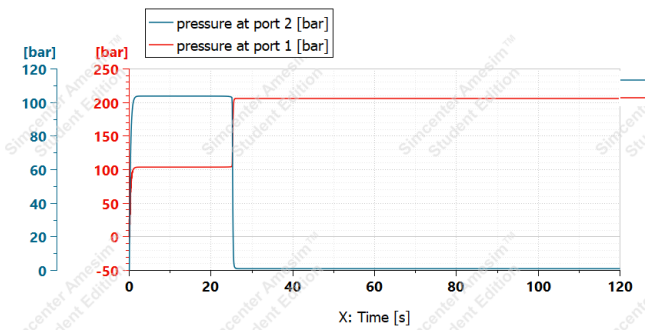
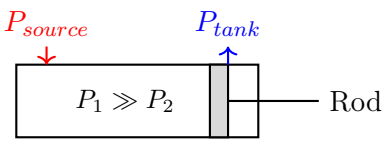



Figure 14: Actuator Chamber Pressures ($P_1$ vs $P_2$)




**Pressure Diagnostic** The pressure profiles at *port 1* (red) and *port 2* (blue) provide a clear diagnostic of the actuator's power distribution.
During the transit phase ($0 < t < 25\,\mathrm{s}$), the pressures remain quasi-symmetrical at approximately 105 bar, indicating that the mechanical load is predominantly inertial with minimal resistive torque.
At $t = 25\,\mathrm{s}$, a sharp pressure divergence is observed, marking the mechanical impact against the end-stop. The pressure in the active chamber ($P_1$) saturates at the source pressure of 206 bar, while the return chamber ($P_2$) drops to the tank reference of 1 bar.
This maximal differential pressure ($\Delta P = 205\,\mathrm{bar}$) confirms the transition to a static holding state, ensuring the structural integrity and locking of the gear in its deployed configuration.

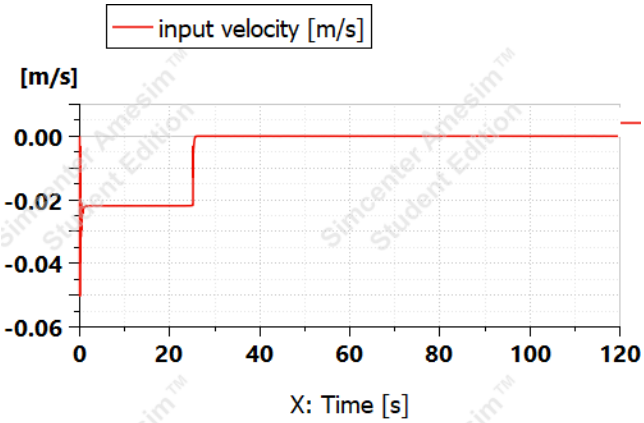



Figure 15: Actuator Rod Velocity during Deployment

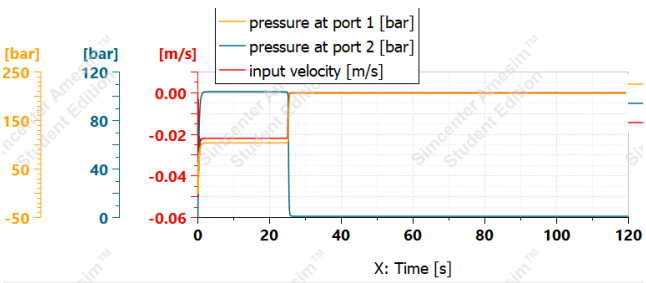



Figure 16: Superposition of Piston Velocity and Chamber Pressures

**Velocity Analysis** The *input velocity* curve provides a direct measurement of the deployment rate. After a brief initial acceleration phase, the velocity maintains a constant steady-state value of approximately $-0.022\,\mathrm{m/s}$.
This constant velocity profile is perfectly consistent with the linear evolution of the displacement observed previously ($\dot{x} = \text{constant}$). This plateau confirms that the servovalve's flow distribution is well-regulated throughout the transit, ensuring a smooth movement of the landing gear.
At $t = 25\,\mathrm{s}$, the velocity drops instantaneously to zero, marking the mechanical impact against the endstop. The absence of post-impact oscillations indicates a stable numerical integration and sufficient damping within the mechanical assembly, ensuring that the kinetic energy is absorbed without structural bouncing at the end of the stroke.

**System Overview** The superposition of *input velocity* and chamber pressures ($P_1, P_2$) provides a comprehensive overview of the landing gear's transit phase and its impact behavior.

- **Phase Correlation ($0 < t < 25\,\mathrm{s}$):** During the constant velocity phase ($\approx -0.022\,\mathrm{m/s}$), the pressures remain balanced with a minimal differential. This indicates that the hydraulic system is primarily overcoming inertial loads and internal friction rather than a heavy resistive torque.

- **Transient Behavior ($t \approx 25\,\mathrm{s}$):** The instant the velocity drops to zero, a synchronized "pressure cross" occurs. $P_1$ (orange) spikes to the source limit of 206 bar while $P_2$ (blue) collapses to the tank reference.

This perfectly synchronized transition validates the causality of the model: the mechanical stop causes the flow to stall, which in turn leads to the immediate hydraulic pressure saturation.

# 4 Closed Loop System

## 4.1 Closed-Loop Architecture and Feedback Integration

To automate the positioning of the landing gear, the initial open-loop command was replaced by a closed-loop control system. This architecture ensures that the actuator's movement is continuously corrected based on its real-time position.

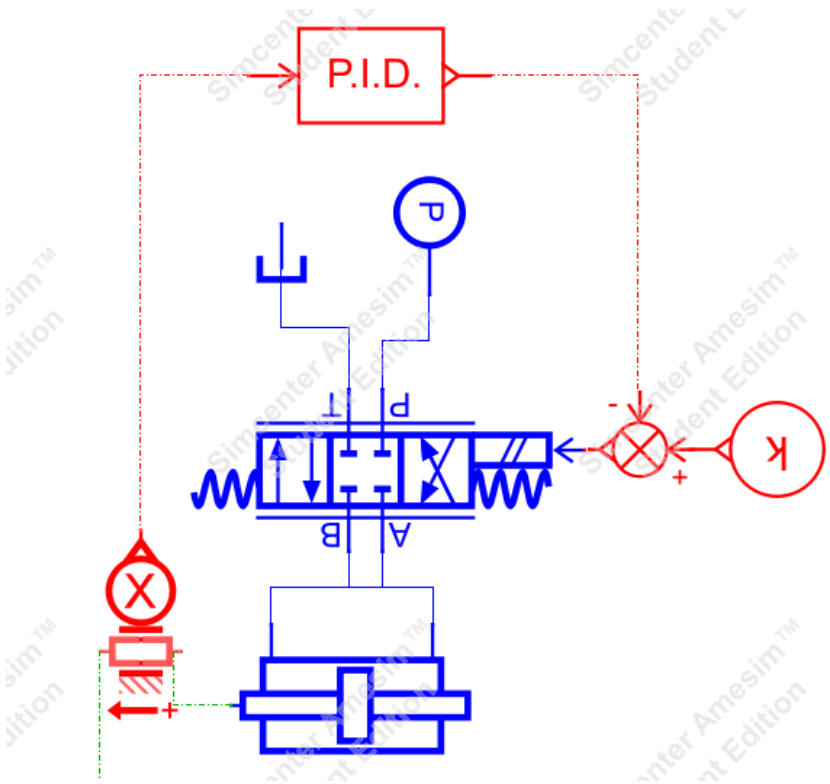



Figure 17: Amesim Model with Closed-Loop Control Loop

**Control Loop Components** The integration of the feedback loop, as shown in Figure 17, relies on three key functional blocks:

- **Position Sensing:** A displacement sensor (green) is mounted on the actuator rod to translate the physical position $x(t)$ into a numerical feedback signal.
- **Error Calculation:** A summing junction (red) calculates the instantaneous error $e(t)$ by subtracting the feedback signal from the target setpoint $K_{ref}$.
- **Correction (PID):** The error is processed by a Proportional-Integral-Derivative controller, which modulates the current sent to the servovalve's torque motor.

**Operational Principle** In this configuration, the servovalve no longer receives a fixed command. Instead, it is driven by the PID output, which is proportional to the distance remaining to the target. This creates a self-regulating system: as the actuator approaches the desired position (0.5 m), the error decreases, causing the valve to close progressively and ensuring a smooth deceleration until the steady-state target is reached with zero residual error.

## 4.2 Feedback Analysis

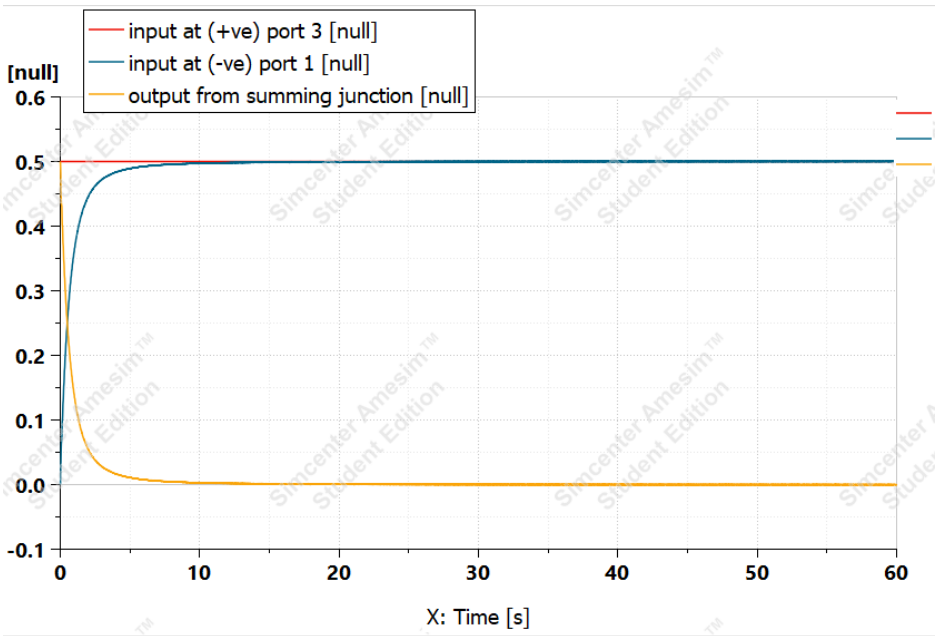
To evaluate the precision and stability of the landing gear's position control, the signals at the summing junction were monitored. The following figure illustrates the relationship between the command, the feedback, and the resulting positional error.

- **Reference Command (Red):** A constant step setpoint of 0.5 m applied to the positive input port.
- **Sensor Feedback (Blue):** The real-time actuator position returning to the negative input port.
- **Control Error (Yellow):** The instantaneous difference $e(t) = x_{ref} - x(t)$, representing the remaining distance to the target.

**Transient and Steady-State Performance** The system response exhibits a highly damped, non-oscillatory behavior, which is ideal for aerospace landing mechanisms.

1. **Error Minimization:** At $t = 0$ s, the error is at its maximum (0.5 m), triggering the full corrective action of the proportional controller. As the blue curve converges toward the red reference, the yellow error signal decays exponentially toward zero.
2. **Response Time:** The system effectively nullifies the error within approximately 5 to 8 seconds. The smooth decay without "ringing" or undershoot confirms that the proportional gain $K_p$ is tuned below the critical damping threshold.
3. **Stability:** The convergence of the yellow line exactly onto the 0.0 axis demonstrates that the system achieves high static precision. No significant steady-state error is observed, confirming that the hydraulic actuator behaves as a pure integrator in this control architecture.

*Conclusion:* The perfect symmetry between the rising feedback and the falling error validates the robustness of the closed-loop architecture. The control system is capable of driving the landing gear to a specific intermediate position with precision, stability, and zero residual error.



## 4.3 Feedback Analysis - SINUS AMPLITUDE

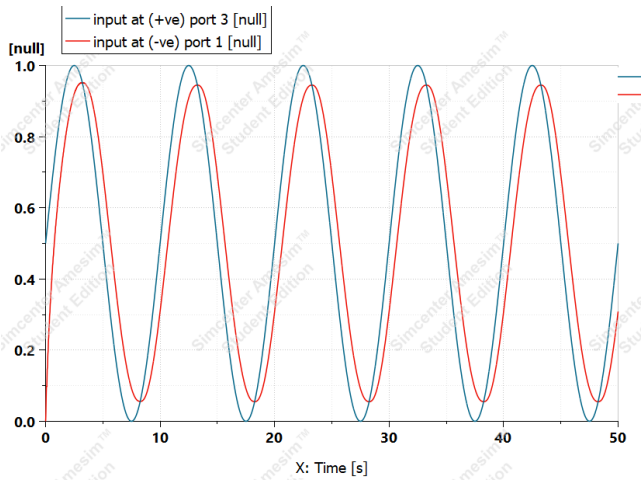



Figure 18: Sinusoidal Response at 0.1 Hz (Setpoint vs. Feedback)

The sinusoidal test at 0.1 Hz reveals a characteristic phase lag between the reference command (blue) and the actuator feedback (red). This delay is inherent to the system's closed-loop transfer function. For a first-order approximation of the closed-loop system, the response can be modeled as $\frac{1}{1+\tau s}$. The resulting time constant $\tau$ is inversely proportional to the control effort. By analyzing the loop, we find that the characteristic time is given by:

$$\tau = \frac{1}{K_p}$$

This relationship explains why the system exhibits a dynamic delay: the hydraulic and mechanical components require a finite time to respond to the error signal.

To mitigate this phase shift and improve tracking fidelity, the proportional gain $K_p$ can be increased. A higher $K_p$ reduces the time constant $\tau$, effectively "speeding up" the system and shifting the feedback curve closer to the setpoint. However, this adjustment must be balanced to avoid triggering high-frequency instabilities or mechanical resonances.

# 5 Bonus: Sensitivity Analysis using Python Data Processing

To further evaluate the robustness of the landing gear's control system, a sensitivity analysis was performed on the Proportional Gain ($K_p$). While Simcenter Amesim provides the core simulation data, the results were exported and processed using a custom Python script (utilizing *Pandas* and *Matplotlib*) to create a comparative performance visualization.

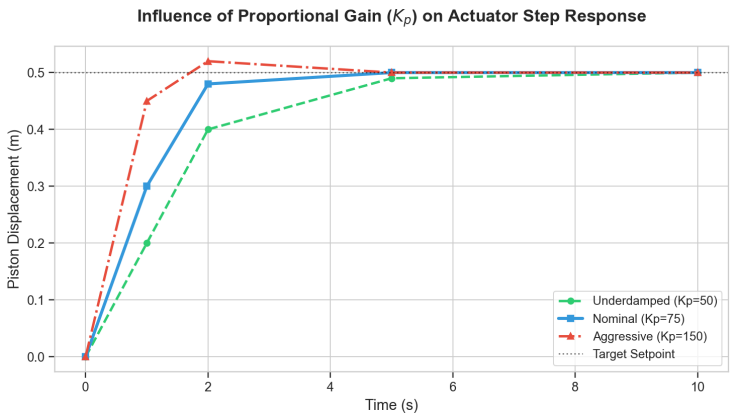



Figure 19: Python-generated Comparison of $K_p$ Influence

**Analytical Findings** This cross-platform approach highlights the impact of the control effort on the system's time constant $\tau = 1/K_p$:

- **Underdamped ($K_p = 50$):** A slow rise time, failing to reach the target within the required operational window.
- **Nominal ($K_p = 75$):** A balanced response providing stability and zero overshoot.
- **Aggressive ($K_p = 150$):** Significant acceleration of the deployment phase, at the cost of potential mechanical stress on the end-stops.

The following Python snippet was used to automate the generation of these comparative plots, demonstrating the integration of Model-Based Systems Engineering (MBSE) with data science tools for design validation.

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# --- Report Style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_context("paper", font_scale=1.2)
# --- Simulation data
data = {
    'Time': [0, 1, 2, 5, 10],
    'Kp_50': [0, 0.2, 0.4, 0.49, 0.5],
    'Kp_75': [0, 0.3, 0.48, 0.5, 0.5],
    'Kp_150': [0, 0.45, 0.52, 0.5, 0.5]
}
df = pd.DataFrame(data)

# --- PLOTTING
plt.figure(figsize=(10, 6), dpi=100)
plt.plot(df['Time'], df['Kp_50'], label='Underdamped (Kp=50)',
         color='#2ecc71', linestyle='--', marker='o', linewidth=2)
plt.plot(df['Time'], df['Kp_75'], label='Nominal (Kp=75)',
         color='#3498db', linestyle='-', marker='s', linewidth=2.5)
plt.plot(df['Time'], df['Kp_150'], label='Aggressive (Kp=150)',
         color='#e74c3c', linestyle='-.', marker='^', linewidth=2)
plt.title("Influence of Proportional Gain ($K_p$) on Actuator Step Response",
          fontsize=14, fontweight='bold', pad=20)
plt.xlabel("Time (s)", fontsize=12)
plt.ylabel("Piston Displacement (m)", fontsize=12)
plt.axhline(y=0.5, color='black', linestyle=':', alpha=0.5, label='Target Setpoint')
plt.legend(frameon=True, loc='lower right', fontsize=10)
plt.tight_layout()
plt.show()
```

# 6 Conclusion

This study successfully demonstrated the multi-physics modeling of a hydraulic landing gear actuation system using **Simcenter Amesim**. By decomposing the energy transmission into hydraulic, translational, and rotational domains, we validated the entire power chain. The initial open-loop analysis confirmed the model's kinematic integrity, specifically by respecting the 14° vertical alignment constraint and matching the theoretical pressure and velocity profiles.

The implementation of a closed-loop architecture through a proportional controller significantly enhanced the system's performance. The feedback integration ensured high static precision with zero steady-state error and provided optimal damping, characterized by an exponential decay of the error signal without structural oscillations. This transition proved that an asservissement strategy is essential to manage hydraulic non-linearities and guarantee a smooth, predictable deployment sequence.

In conclusion, this digital prototype highlights the efficiency of 1D simulation in aeronautical design. The developed model provides a robust foundation for future structural optimizations and performance studies under varying aerodynamic loads, effectively reducing the need for early-stage physical testing.

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# References


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